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PAPER OF PT4-ACC-24CAI0104-B2024-SEM2-CLONE EXAM

Q1. ACC-24CAI0104-CODING-5MARKS-PT4-B2024-AI-ARRAY

Running Comparison

Alice and Bob recently got into running, and decided to compare how much they ran on different days.

They each ran for N days — on the ith day, Alice ran *Ai* meters and Bob ran *Bi* meters.

On each day,

- Alice is *unhappy* if Bob ran **strictly more than twice** her distance, and *happy* otherwise.
- Similarly, Bob is *unhappy* if Alice ran **strictly more than twice** his distance, and *happy* otherwise.

For example, on a given day

- If Alice ran 200 meters and Bob ran 500, Alice would be unhappy but Bob would be happy.
- If Alice ran 500 meters and Bob ran 240, Alice would be happy and Bob would be unhappy.
- If Alice ran 300 meters and Bob ran 500, both Alice and Bob would be happy.

On how many of the N

N days were **both** Alice and Bob happy?

Input Format

- The first line of input will contain a single integer *T*, denoting the number of test cases.
- Each test case consists of three lines of input.

The first line of each test case contains a single integer *N* — the number of days.

The second line of each test case contains *N* space-separated integers A1,A2,…,AN — the distances run by Alice on the *N* days.

- The third line of each test case contains *N* space-separated integers B1,B2,…,BN — the distances run by Bob on the *N* days.

Output Format

For each test case, output on a new line the number of days when **both** Alice and Bob were happy.

Constraints

- $1 \le T \le 1000$
- $1 \le N \le 100$
- $1 \le A_i, B_i \le 10^5$

Sample Input 1:

4

3

100 200 300

300 200 100

4

1000 1000 1000 1000

400 500 600 1200

4

800 399 1400 532

2053 2300 3400 23

5

800 850 900 950 1000

600 800 1000 1200 1400

sample output1:

1

3

0

5

Explanation:

Test case 1

Alice is unhappy on the first day, since she ran 100 meters but Bob ran 300; and 300 is more than twice of 100.

Similarly, Bob is unhappy on the third day.

Both Alice and Bob are happy on the second day, so the answer is 1

Test case 2

Bob is unhappy on day 1 and happy on every other day, while Alice is happy on every day.

So, they're both happy on three days — the second, third, and fourth.

Test case 3

Alice is unhappy on days 1,2,3 and Bob is unhappy on day 4.

Test case 4

Alice and Bob are both happy on all five days.

ANSWER:

Test case 1 :

Input:
4
3
100 200 300
300 200 100
4
1000 1000 1000 1000
400 500 600 1200
4
800 399 1400 532
2053 2300 3400 23
5
800 850 900 950 1000
600 800 1000 1200 1400
Output:
1
3
0
5

Q2. ACC-24CAI0104-CODING-5MARKS-PT4-B2024-AI-ARRAY

Largest and Second Largest

You are given an array A of N integers.

Find the **maximum** sum of **two distinct** integers in the array.

Note: It is guaranteed that there exist at least two distinct integers in the array.

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases.
- Each test case consists of multiple lines of input.

The first line of each test case contains single integer N — the size of the array.

The next line contains N space-separated integers, denoting the array A.

Output Format

For each test case, output on a new line, the maximum sum of two distinct integers in the array.

Constraints

- $1 \le T \le 1000$
- $2 \le N \le 10^5$
- $1 \le A_i \le 1000$
- The sum of N over all test cases does not exceed $2 \cdot 10^5$

Sample 1:

Input:

4

3

4 1 6

7

3 7 2 1 1 5 3

5

8 2 9 4 9

2

1 2

Output:

10

12

17

3

Explanation:

Test case 1

The maximum sum of two distinct elements is 4+6=10 .

Test case 2

The maximum sum of two distinct elements is 7+5=12.

Test case 3

The maximum sum of two distinct elements is 8+9=17.

Test case 4

The maximum sum of two distinct elements is 1+2=3.

ANSWER:

Test case 1 :

Input:
4
3
4 1 6
7
3 7 2 1 1 5 3

5
8 2 9 4 9
2
1 2
Output:
10
12
17
3

Q3. ACC-24CAI0104-CODING-5MARKS-PT4-B2024-AI-ARRAY

Difficulty Rating Order

Our Chef has some students in his coding class who are practicing problems. Given the difficulty of the problems that the students have solved in order, help the Chef identify if they are solving them in non-decreasing order of difficulty. Non-decreasing means that the values in an array is either increasing or remaining the same, but not decreasing. That is, the students should not solve a problem with difficulty d1, and then later a problem with difficulty d2, where d1>d2.

Output “Yes” if the problems are attempted in non-decreasing order of difficulty rating and “No” if not.

Input Format

- The first line of input will contain a single integer T, denoting the number of test cases. The description of the test cases follows.
- Each test case consists of 2 lines of input.

The first line contains a single integer N, the number of problems solved by the students

The second line contains N space-separate integers, the difficulty ratings of the problems attempted by the students in order.

Output Format

- For each test case, output in a new line “Yes” if the problems are attempted in non-decreasing order of difficulty rating and “No” if not. The output should be printed without the quotes.

Each letter of the output may be printed in either lowercase or uppercase. For example, the strings yes, YeS, and YES will all be treated as equivalent.

Constraints

- $1 \le T \le 100$
- $2 \le N \le 100$
- $1 \le$ difficulty of each problem ≤ 5000

Sample 1:

Input:

4

3

1 2 3

3

1 1 2

5

100 200 300 400 350

5

1000 2000 5000 3000 1000

Output:

Yes

Yes

No

No

Explanation:

Test case 1

$1 \le 2 \le 3$. The students have solved problems in increasing order, and so the answer is "Yes".

Test case 2

1≤1≤2. The students have solved problems in non-decreasing order, and so the answer is "Yes".

Test case 3

400>350, but the students have solved a 400-difficulty level problem before solving a 350-difficulty problem. The students have not solved problems in non-decreasing order, and so the answer is "No".

Test case 4

5000>3000, but the students have solved a 5000-difficulty level problem before solving a 3000-difficulty problem. The students have not solved problems in non-decreasing order, and so the answer is "No".

ANSWER:

Test case 1 :

Input:
4
3
1 2 3
3
1 1 2
5
100 200 300 400 350
5
1000 2000 5000 3000 1000
Output:
Yes
Yes
No
No

Q4. ACC-24CAI0104-CODING-5MARKS-PT4-B2024-AI-STRING

Add One

You are given a large number N. You need to print the number $N+1$.

Note: The number is very large and it will not fit in standard integer data type. You have to take the input as String and then manipulate the digits to convert it to $N+1$.

Input Format

- The first line of the input contains a single integer T - the number of test cases. The description of T test cases follows.
- The first line of each test case contains a single integer N.

Output Format

- For each test case, print a single line containing one integer - the number $N+1$.

Constraints

- $1 \le T \le 100$
- $1 \le N \le 10200000-1$
- the sum of the number of digits of all N in a single test file does not exceed $4 \cdot 105$.

Subtasks

Subtask #1 (30 points):

- each digit of the number N is at most 8.

Subtask #2 (70 points): original constraints

Sample 1:

Input:

6

99

17

1

599

10000000000000000000

549843954323494990404

Output:

100

18

2

600

10000000000000000001

549843954323494990405

Explanation:

Example case 1: N=99 so N+1=100.

Example case 2: N=17 so N+1=18.

Example case 3: N=1 so N+1=2.

Example case 4: N=599 so N+1=600.

Example case 5: N=10000000000000000000 so N+1=10000000000000000001.

Example case 6: N=549843954323494990404 so N+1=549843954323494990405.

ANSWER:

Test case 1 :

Input:
6
99
17
1
599
10000000000000000000
549843954323494990404
Output:

100
18
2
600
10000000000000000001
549843954323494990405

Q5. ACC-24CAI0104-CODING-5MARKS-PT4-B2024-AI-STRING

Chef and Happy String

Chef has a string S with him. Chef is happy if the string contains a contiguous substring of length strictly greater than 2 in which all its characters are vowels.

Determine whether Chef is happy or not.

Note that, in english alphabet, vowels are a, e, i, o, and u.

Input Format

First line will contain T, number of test cases. Then the test cases follow.

Each test case contains of a single line of input, a string S.

Output Format

For each test case, if Chef is happy, print HAPPY else print SAD.

You may print each character of the string in uppercase or lowercase (for example, the strings hAppY, Happy, haPpY, and HAPPY will all be treated as identical).

Constraints

$1 \le T \le 1000$

$3 \le |S| \le 1000$, where $|S|$ is the length of S.

S will only contain lowercase English letters.

Sample 1:

Input:

4

aeiou

abxy

aebcdefghij

abcdeeafg

Output:

Happy

Sad

Sad

Happy

Explanation:

Test case 1

Since the string aeiou is a contiguous substring and consists of all vowels and has a length >2, Chef is happy.

Test case 2

Since none of the contiguous substrings of the string consist of all vowels and have a length >2, Chef is sad.

Test case 3

Since none of the contiguous substrings of the string consist of all vowels and have a length >2, Chef is sad.

Test case 4

Since the string eea is a contiguous substring and consists of all vowels and has a length >2, Chef is happy.

ANSWER:

Test case 1 :

Input:
4
aeiou
abxy
aebcdefghij
abcdeeafg
Output:
Happy
Sad
Sad
Happy

Q6. ACC-24CAI0104-CODING-5MARKS-PT4-B2024-AI-STRING

Blobby Volley Scores

Alice and Bob are playing a game of Blobby Volley. In this game, in each turn, one player is the server and the other player is the receiver. Initially, Alice is the server, and Bob is the receiver.

If the server wins the point in this turn, their score increases by 1, and they remain as the server for the next turn.

But if the receiver wins the point in this turn, their score does not increase. But they become the server in the next turn.

In other words, your score increases only when you win a point when you are the server.

Please see the Sample Inputs and Explanation for more detailed explanation.

They start with a score of each, and play *N* turns. The winner of each of those hands is given to you as a string consisting of 'A's and 'B's. 'A' denoting that Alice won that point, and 'B' denoting that Bob won that point. Your job is the find the score of both of them after the *N* turns.

Input Format

- The first line of input will contain a single integer *T*, denoting the number of test cases.
- Each test case consists of two lines of input.

The first line of each test case contains one integer *N* — the number of turns.

The line contains a string *S* of length *N*.

If the *ith* character of this string is 'A', then Alice won that point.

If the *ith* character of this string is 'B', then Bob won that point.

Output Format

For each test case, output on a new line, two space-separated integers - Alice's final score, and Bob's final score.

Constraints

- $1 \leq T \leq 1000$
- $1 \leq N \leq 1000$
- Length of $|S| = N$
- *S* consists only of the characters 'A' and 'B'.

Sample 1:

Input:

4

3

AAA

4

BBBB

5

ABABB

5

BABAB

Output:

3 0

0 3

1 1

0 0

Explanation:

Test case 1: The given string is "AAA".

- Score is 0, 0.
- Turn 1. Alice is the server, and Bob is the receiver. Alice wins.
- Score is 1, 0.
- Turn 2. Alice is the server, and Bob is the receiver. Alice wins.
- Score is 2, 0.
- Turn 3. Alice is the server, and Bob is the receiver. Alice wins.
- Score is 3, 0.

This is the final score, and so the output is 3 0.

Test case 2: The given string is "ABABB".

- Score is 0, 0.
- Turn 1. Alice is the server, and Bob is the receiver. Alice wins.
- Score is 1, 0.
- Turn 2. Alice is the server, and Bob is the receiver. Bob wins.
- Score is 1, 0. [Note that Bob's score doesn't increase]
- Turn 3. Bob is the server, and Alice is the receiver. Alice wins.
- Score is 1, 0. [Note that Alice's score doesn't increase]
- Turn 4. Alice is the server, and Bob is the receiver. Bob wins.
- Score is 1, 0. [Note that Bob's score doesn't increase]
- Turn 5. Bob is the server, and Alice is the receiver. Bob wins.
- Score is 1, 1.

This is the final score, and so the output is 1 1.

Test case 3: The given string is "BABAB".

- Score is 0, 0.
- Turn 1. Alice is the server, and Bob is the receiver. Bob wins.
- Score is 0, 0.
- Turn 2. Bob is the server, and Alice is the receiver. Alice wins.
- Score is 0, 0.
- Turn 3. Alice is the server, and Bob is the receiver. Bob wins.

- Score is 0, 0.
- Turn 4. Bob is the server, and Alice is the receiver. Alice wins.
- Score is 0, 0.
- Turn 5. Alice is the server, and Bob is the receiver. Bob wins.
- Score is 0, 0.

This is the final score, and so the output is 0 0.

ANSWER:

Test case 1 :

Input:
4
3
AAA
4
BBBB
5
ABABB
5
BABAB
Output:
3 0
0 3
1 1
0 0